

3

Current Control

In electrical drives, the controlled quantity is ultimately the motor torque, and intermediately currents and possibly the motor flux. In power electronic applications where the power converter is connected to the grid with the purpose of injecting or controlling currents in the grid, the currents is the prime target for the control algorithms. In many other applications, such as battery charging, lighting, active anti corrosion equipment etc. it is generally the current that is the controlled quantity.

This chapter deals with current control methods suitable for power electronic converters with idealized loads, i.e. with loads consisting of a resistance (R), inductance (L) and counter emf (e) in series. The methods developed for such loads are easily transferred and adapted to more complex loads such as electrical machines, but the theoretical derivation of the control algorithms are more transparent and didactic with the idealized loads.

3.1 Discrete control

In this chapter there will be three different assumptions made regarding the signal processing speed of the current control computer:

Fast computer, fix sampling interval. This means that the computer is able to provide a voltage reference for a sampling interval that is based on measurements sampled in the beginning of the sampling interval. This is relevant for analogue controllers, but generally not for digital since the processing speed is finite even with the fastest processors available today.

Slow computer, fix sampling interval. This means that the computer provides a voltage reference that is based on the measurement of currents that is made in the beginning of the previous sampling interval.

Fast computer, continuous sampling. This means that the computer samples and provides switch position selection as fast as it can. This applies to direct control methods, i.e. the switch positions are determined directly from the current errors, without going through an intermediate current control algorithm followed by a modulator.

Figure 3.1 illustrates the three different ways of using the control computer. Note that one specific computer can be used in all three different ways, depending on the sampling time and the dynamics of the application. The slow computer is usually the reality with carrier wave modulation, since the sampling interval usually is in the range of [100...500] microseconds, and sampling, estimation of non-measured quantities and calculation of the

control algorithms take several 10's of milliseconds, i.e. most of the sampling interval is passed before the voltage reference is finally calculated. As will be shown later, there are intelligent prediction methods that partially compensates for the slow calculation speed.

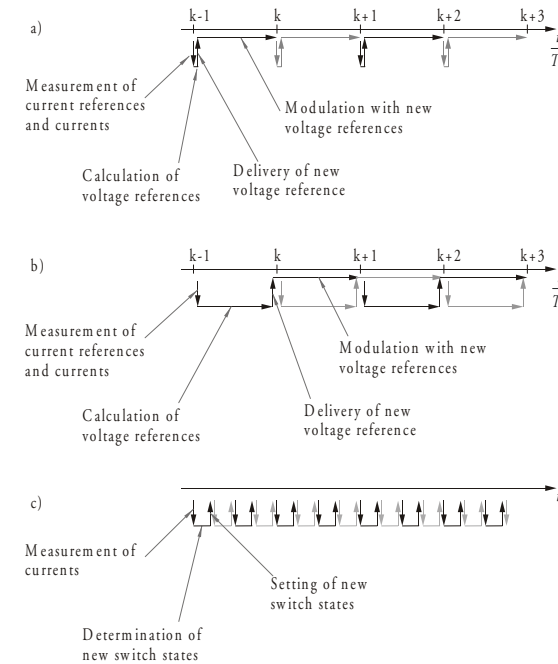


Figure 3.1: Calculation speed related to sampling time instants $[k, k+1, k+2, \dots]$ for a) a fast computer, b) a slow computer and with continuous sampling c).

When deriving current control algorithms, both the assumption of a fast computer as well as that of a slow computer will be used, and the consequence for the design of the controller parameters will be highlighted. The continuous sampling alternative is handled differently since it does not involve any current control algorithm.

3.2 Single-phase load model

The single phase load model is a resistance (R), an inductance (L) and an induced emf (e) in series according to Figure 3.2.

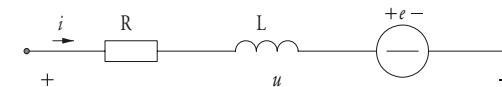


Figure 3.2: A generic 1 phase load.

The equation representing the 1-phase load is given in (3.1).

$$u = R \cdot i + L \cdot \frac{di}{dt} + e \quad (3.1)$$

Note that the parameters R and L in the 1-phase load are considered to be constant, i.e. no temperature dependence in the resistor and no saturation dependence in the inductor.

3.3 3-phase load model

The three phase load model consists of a symmetric resistive-inductive (R-L) load in series with a symmetric three phase voltage vector ($\vec{e}$).

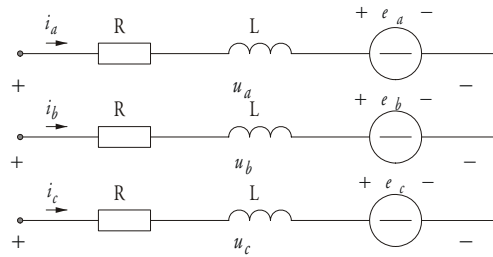


Figure 3.3: 3-phase load.

The equations describing the three phase load according to Figure 3.1 can be described as three scalar equations of one vector equation according to:

$$\begin{aligned} & \sqrt{\frac{2}{3}} \left(u_a = R \cdot i_a + L \cdot \frac{di_a}{dt} + e_a \right) \\ & \sqrt{\frac{2}{3}} \cdot e^{j\frac{2\pi}{3}} \cdot \left(u_b = R \cdot i_b + L \cdot \frac{di_b}{dt} + e_b \right) \\ & + \sqrt{\frac{2}{3}} \cdot e^{j\frac{4\pi}{3}} \cdot \left(u_c = R \cdot i_c + L \cdot \frac{di_c}{dt} + e_c \right) \\ & \vec{u} = R \cdot \vec{i} + L \cdot \frac{d\vec{i}}{dt} + \vec{e} \end{aligned} \quad (3.2)$$

With a symmetric load voltage $\{e_a, e_b, e_c\}$ according to equation (3.3), the load voltage vector can be derived according to equation (3.4).

$$\begin{cases} e_a = \hat{e} \cdot \cos(\omega \cdot t) \\ e_b = \hat{e} \cdot \cos\left(\omega \cdot t - \frac{2\pi}{3}\right) \\ e_c = \hat{e} \cdot \cos\left(\omega \cdot t - \frac{4\pi}{3}\right) \end{cases} \quad (3.3)$$

$$\begin{aligned} \vec{e} &= \sqrt{\frac{2}{3}} \cdot \left(e_a + e_b \cdot e^{j\frac{2\pi}{3}} + e_c \cdot e^{j\frac{4\pi}{3}} \right) = \\ &= \sqrt{\frac{2}{3}} \cdot \hat{e} \cdot \left(\cos(\omega \cdot t) + \cos\left(\omega \cdot t - \frac{2\pi}{3}\right) \cdot \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) + \cos\left(\omega \cdot t - \frac{4\pi}{3}\right) \cdot \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \right) = \\ &= \sqrt{\frac{2}{3}} \cdot \hat{e} \cdot \left(\cos(\omega \cdot t) + \left(\cos(\omega \cdot t) \cdot \cos\left(\frac{2\pi}{3}\right) + \sin(\omega \cdot t) \cdot \sin\left(\frac{2\pi}{3}\right) \right) \cdot \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \right. \\ &\quad \left. + \left(\cos(\omega \cdot t) \cdot \cos\left(\frac{4\pi}{3}\right) + \sin(\omega \cdot t) \cdot \sin\left(\frac{4\pi}{3}\right) \right) \cdot \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \right) = \\ &= \sqrt{\frac{2}{3}} \cdot \hat{e} \cdot \left(\cos(\omega \cdot t) \cdot \left(1 + \frac{1}{4} + \frac{1}{4}\right) + j \cdot \sin(\omega \cdot t) \cdot \left(\frac{3}{4} + \frac{3}{4}\right) \right) = \\ &= \sqrt{\frac{3}{2}} \cdot \hat{e} \cdot (\cos(\omega \cdot t) + j \cdot \sin(\omega \cdot t)) = E \cdot e^{j\omega t} \end{aligned} \quad (3.4)$$

Note that in equation (3.4) E is the rms value of the phase-to-phase voltage. The load voltage is thus a vector with constant length E and constant rotating speed ω . It is useful to arrange a reference frame to the integral of this voltage vector, denoted $\vec{\psi}$ according to equation (3.5) and Figure 3.4.

$$\begin{aligned} \vec{\psi} &= \int_0^t \vec{e} \cdot dt = \int_0^t E \cdot e^{j\omega \cdot t} dt = \frac{\vec{e}}{j \cdot \omega} = \\ &= \frac{E}{\omega} e^{j\left(\omega \cdot t - \frac{\pi}{2}\right)} \end{aligned} \quad (3.5)$$

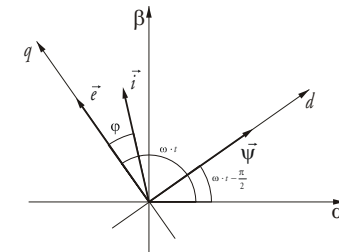


Figure 3.4: Rotating reference frame defined by the integral of the load emf vector.

The voltage equation can be expressed in the (d,q) reference frame aligned with the integral of the load voltage vector.

$$\vec{u} = R \cdot \vec{i} + L \cdot \frac{d\vec{i}}{dt} + j \cdot \omega \cdot L \cdot \vec{i} + \vec{e} \quad (3.6)$$

The active power delivered to the load, expressed in the (d,q) quantities is give by equation (3.7).

$$\begin{aligned} \vec{u} &= R \cdot \vec{i} + L \cdot \frac{d\vec{i}}{dt} + j \cdot \omega \cdot L \cdot \vec{i} + \vec{e} \\ p(t) &= \text{Re}\{\vec{u} \cdot \vec{i}^*\} = \text{Re}\left\{R \cdot \vec{i} \cdot \vec{i}^* + L \cdot \frac{d\vec{i}}{dt} \cdot \vec{i}^* + j \cdot \omega \cdot L \cdot \vec{i} \cdot \vec{i}^* + \vec{e} \cdot \vec{i}^*\right\} = \\ &= \text{Re}\left\{R \cdot \vec{i} \cdot \vec{i}^* + L \cdot \frac{d\vec{i}}{dt} \cdot \vec{i}^* + j \cdot \omega \cdot L \cdot \vec{i} \cdot \vec{i}^* + \vec{e} \cdot \vec{i}^*\right\} = \\ &= \underbrace{R i_d^2 + R i_q^2}_1 + \underbrace{L \frac{di_d}{dt} i_d + L \frac{di_q}{dt} i_q}_2 + \underbrace{e_q i_q}_3 \end{aligned} \quad (3.7)$$

The first term (1) in equation (3.7) is the losses in the load resistance. The second term (2) is the power delivered to the inductance upon a change of the load current. Finally the third term (3) is the power absorbed by the load emf $\vec{e}$ that by definition is oriented to the q -axis.

Note that the active power absorbed by the load emf $\vec{e}$ at symmetrical emf's according to equation (3.3), and with stationary symmetrical sinusoidal load currents is:

$$p(t) = E \cdot \sqrt{\frac{3}{2}} \cdot |\vec{i}| \cdot \cos(\varphi) = \sqrt{3} \cdot E \cdot I_{rms, phase} \cdot \cos(\varphi) \quad (3.8)$$

where $I_{rms, phase}$ is the rms value of the phase current. Note that equation (3.8) is equivalent to the conventional power expression in stationary symmetrical sinusoidal three phase systems. Note also that not equation (3.7), that is the instantaneous power without requirement of sinusoidal stationarity, nor equation (3.8), that requires sinusoidal stationarity, has any term corresponding to the conventional expression for reactive power. The expression “instantaneous reactive power” that sometimes is heard in the power engineering community is misleading, there is no such thing as instantaneous reactive power!

3.4 Sampled current control of a generic single phase load with a fast computer

It is assumed that the current is given at the sampling time instant as $i(k)$ and the current reference at the same instant $i^*(k)$. Furthermore, it is assumed that the current error $(i^*(k) -$

$i(k)$) will be eliminated during the next sampling interval, i.e. during the interval $[k, k+1]$. The voltage needed during this interval can thus be derived from the voltage equation (3.1) by integration over one sampling interval T_s according to equation (3.9).

$$\begin{aligned} \frac{\int_{k \cdot T_s}^{(k+1)T_s} u \cdot dt}{T_s} &= \frac{R \cdot \int_{k \cdot T_s}^{(k+1)T_s} i \cdot dt + L \cdot \int_{k \cdot T_s}^{(k+1)T_s} \frac{di}{dt} \cdot dt + \int_{k \cdot T_s}^{(k+1)T_s} e \cdot dt}{T_s} = \\ &= \bar{u}(k, k+1) = R \cdot \bar{i}(k, k+1) + L \cdot \frac{i(k+1) - i(k)}{T_s} + \bar{e}(k, k+1) \end{aligned} \quad (3.9)$$

To convert the “average voltage”-equation (3.9) into a current control algorithm some assumptions have to be made:

$$\begin{aligned} \bar{u}(k, k+1) &= u^*(k) & (a) \\ i(k+1) &= i^*(k) & (b) \\ \bar{i}(k, k+1) &= \frac{i^*(k) + i(k)}{2} & (c) \\ \bar{e}(k, k+1) &= e(k) & (d) \\ i(k) &= \sum_{n=0}^{n=k-1} (i^*(n) - i(n)) & (e) \end{aligned} \quad (3.10)$$

Assumption (a) is straight forward, the reference voltage for the coming period should be the voltage needed to achieve the desired current change. Assumption (b) is an assumption of “dead beat” current control, i.e. the entire current error is eliminated in one sampling interval. Assumption (c) is an approximation, and is not always true, since the trajectory of the current will depend on the relation between the load emf e , the dc link voltage U_{dc} and the switching frequency. However, the assumption (c) is true in average, and furthermore it is related to the resistive voltage drop that normally is relatively small, thus as small error in this assumption is not expected to distort the current control performance significantly. The assumption (d) is usually true since the sampling frequency usually is several orders of magnitude faster than the dynamics of the load emf. One example is a DC machine where the emf corresponds to the induced voltage due to speed, which usually changes in the [10...100] millisecond range, whereas the sampling interval is in the [100-1000] microsecond range. The assumption (e) is a consequence of the “dead beat” assumption (b). The current at instant k is the sum of all changes of currents until the last change depicted in the previous sampling instant. With all these assumptions a current controller can be derived from the “average voltage” equation (3.9) as:

$$\begin{aligned}
u^*(k) &= R \cdot \frac{i^*(k) + i(k)}{2} + L \cdot \frac{i^*(k) - i(k)}{T_s} + e(k) = \\
&= R \cdot \frac{i^*(k) - i(k)}{2} + R \cdot i(k) + L \cdot \frac{i^*(k) - i(k)}{T_s} + e(k) = \\
&= \left(\frac{L}{T_s} + \frac{R}{2} \right) (i^*(k) - i(k)) + R \cdot \sum_{n=0}^{n=k-1} (i^*(n) - i(n)) + e(k) = \\
&= \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left(\underbrace{(i^*(k) - i(k))}_{\text{Proportional}} + \underbrace{\left(\frac{T_s}{\left(\frac{L}{R} + \frac{T_s}{2} \right)} \cdot \sum_{n=0}^{n=k-1} (i^*(n) - i(n)) \right)}_{\text{Integral}} \right) + \underbrace{e(k)}_{\text{Feed forward}}
\end{aligned} \tag{3.11}$$

Note that a “PIE” current controller has been derived, with both gain and integral time constant directly related to the circuit parameters, and with a feed forward term directly related to the emf of the load. These properties are useful since it is now possible to determine the controller sensitivity to changes of a particular parameter, e.g. saturation dependent inductances and temperature dependent resistances.

3.5 Sampled current control of a generic single phase load with a slow computer

The case with a slow computer corresponds to a second order system, with a delay of one sample interval between the calculation of an output voltage set point and the actual use of that set point in the modulation. The most common way to get around this problem is to use a prediction of the measured current instead of the real measured current. The prediction of the current at sample instant $\{k\}$ is based on the measured current at time instant $\{k-1\}$ extrapolated with the expected change of the current due to the known applied voltage during the interval $\{k-1, k\}$, i.e. $u^*(k-1)$. This can be done in at least two different ways:

- A) The expected current change can be based on the applied voltage during the interval $\{k-1, k\}$, i.e. $u^*(k-1)$, the estimated resistive voltage drop and the estimated load emf, by using equation (3.1). The local current change is thus calculated, but no state is remembered for future use. The advantage of this method is the accuracy if all the input variables (voltage reference, resistive voltage drop and induced emf) are known. On the other hand, in the case that these variables are not known well enough, the errors in these may excite control errors.
- B) A simplified load model can be made that does not include any induced emf, only an inductance and a resistance. In this simplified model the state of the current will be remembered, and it will be wrong, but the change of the current will be correct when the input signal, the voltage reference is changing. Since it is the state that is desired, such a simplified state model is enough.

Both these methods are variants of the same idea, often referred to as a “Smith predictor”. The circuit model in the single phase load model can be derived as a discrete model since the controller is discrete, with the emf omitted:

$$\begin{aligned}
u &= R \cdot i + L \cdot \frac{di}{dt} \quad (\text{NOTE! No emf modelled}) \\
u^*(k) &= u(k, k+1) = R \cdot i_{sp}(k) + \frac{L}{T_s} \cdot (i_{sp}(k+1) - i_{sp}(k))
\end{aligned}$$

or, in the discrete domain;

$$\begin{aligned}
u &= R \cdot i_{sp} + \frac{L}{T_s} \cdot (z-1) \cdot i_{sp} \\
\frac{i_{sp}}{u} &= \frac{1}{R + \frac{L}{T_s} \cdot (z-1)} = \frac{1}{\frac{L}{T_s} \cdot z + (R - L/T_s)} = \frac{T_s}{L} \cdot \frac{1}{z - (1 - R \cdot T_s / L)}
\end{aligned} \tag{3.12}$$

or, again in the time domain;

$$i_{sp}(k+1) = i_{sp}(k) \cdot (1 - R \cdot T_s / L) + u^*(k) \cdot \frac{T_s}{L}$$

With the transfer function of equation 3.12, a “dummy” current i_{smith} is created, that does not have the same stationary value as the real current of the circuit that we are really controlling, but has the same dynamics. In stationarity, the dummy current i_{smith} will reach a level determined by the actual voltage and the resistance only. However, when any change is made to the voltage reference there will be a change in the dummy current i_{smith} that will be identical to whatever it would have been even if the induced emf was represented in the model. Since we are only interested in the change of the current to predict the actual circuit current based on the latest measurement and the smith-prediction, this dummy model is enough. The current estimate to be used by the controller of the “slow computer” is thus finally calculated as:

$$\hat{i}(k+1) = i(k) + (i_{sp}(k+1) - i_{sp}(k))$$

The accuracy of the current prediction is of course dependent on the accuracy of the parameters and the estimate of the induced voltage used by the controller. A simple sensitivity analysis is presented in the next section. The current controller algorithm when using a slow computer is thus given by equation (3.13).

$$u^*(k) = \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left(i^*(k) - \hat{i}(k) \right) + \frac{T_s}{\left(\frac{L}{R} + \frac{T_s}{2} \right)} \cdot \sum_{n=0}^{n=k-1} (i^*(n) - \hat{i}(n)) + \hat{e}(k) \tag{3.13}$$

The quality of the current prediction depends on application. With a single phase load like the one used in this chapter, it is possible to measure e.g. the emf e , and the inductor

parameters are usually rather easy to measure. When controlling e.g. a DC machine, the emf is not available for measurement and the estimate is of course more difficult to make. It remains to be determined in the specific application what methods can be used, and with what accuracy, to estimate circuit parameters and the emf.

Sampled current controller applied to a 2 quadrant converter.

The current controllers derived in the previous sections will now be applied to study some different behaviors depending on limitations in the converter. A circuit with the controller, a 2-quadrant converter and the one phase load is shown in Figure 3.5.

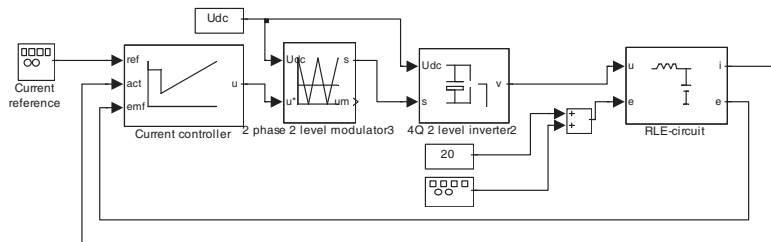


Figure 3.5: 2 quadrant DC link converter.

Assume a load circuit and a 2-quadrant DC converter with parameters according to:

$$\begin{aligned} L &= 10 \text{ mH} \\ R &= 1 \text{ } \Omega \\ Ts &= 0.5 \text{ ms} \\ U_{dc} &= 100 \text{ V} \end{aligned}$$

To evaluate the performance of the developed current controllers, a series of simulations are made. In all cases a series of current reference steps between $\pm 5 \text{ A}$ are used. The following situations are simulated:

1. The current controller has the right set of parameters, i.e. a perfect measurement of the load parameters and an as perfect measurement of the load emf is used. The controller is of the "fast computer" type. Three different cases are simulated, each with a unique emf e to show the dynamic performance at different emf's.
2. In the other cases the current controller is of the slow computer type, and the controller parameters are deliberately different from those of the load, to show the performance of a mistuned current controller. Six different cases are simulated; three with $R_{controller} = [0.5 \text{ } 1.0 \text{ } 2.0]R$ and $L_{controller} = L$, and three with $L_{controller} = [0.9 \text{ } 1.0 \text{ } 1.1]L$ and $R_{controller} = R$. The load emf is $e = 30 \text{ V}$ in all cases.

Figure 3.6 shows the step response for situation 1 above. Note the difference in raise time and fall time at the three different emf's. Note how the current controller takes more than

one sample interval to settle the step response. The controller is equipped with anti windup function. Would it be possible to have the negative half period of the current in case that the emf $e = 0 \text{ V}$?

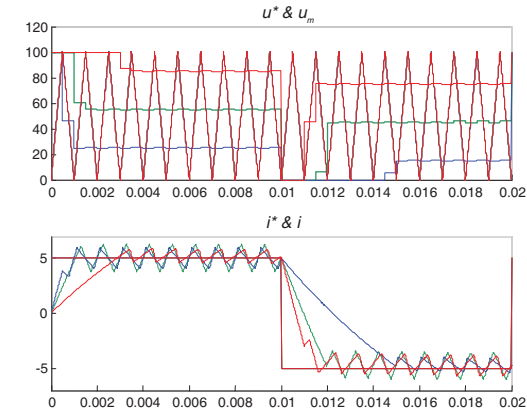


Figure 3.6: Load current step response. Upper diagram with modulating wave and references with $e = [20, 50, 80] \text{ V}$. Lower diagram with current reference and current.

Figure 3.7 shows the current step response for situation II above, with the estimated load resistance tuned to 50%, 100% and 200% respectively. Note the small difference between the three cases, indicating a low sensitivity to variations in the load resistance.

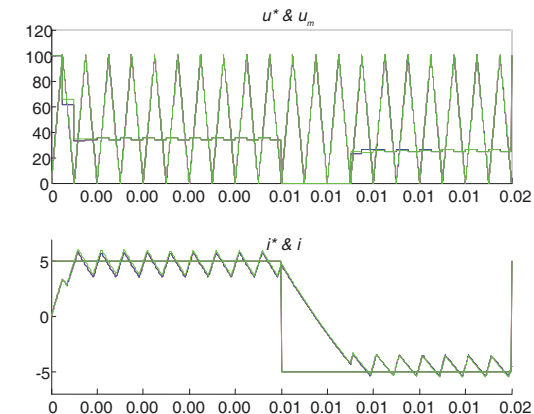


Figure 3.7: Current step response with a "slow computer" current controller and mistuned resistance with [50%, 100%, 200%] of the load resistance used in the controller.

Figure 3.8 shows the current step response for situation II above, with the estimated load inductance tuned to 80%, 100% and 120% respectively.

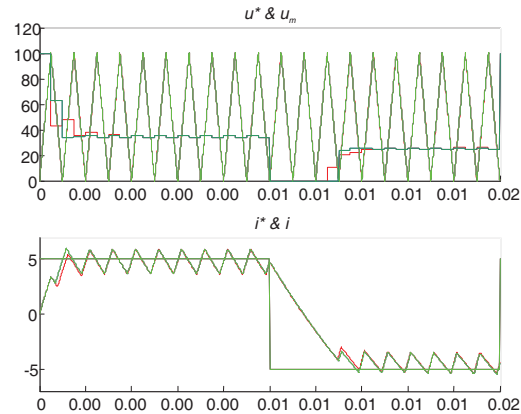


Figure 3.8: Current step response with a "slow computer" current controller and mistuned inductance with [80%, 100%, 120%] of the load inductance used in the controller.

Sampled current controller applied to a 4-quadrant converter.

The current controllers derived in the previous sections will now be applied to study some different behaviors depending on limitations in the converter. A circuit with the controller, a 2-quadrant converter and the one phase load is shown in Figure 3.9.

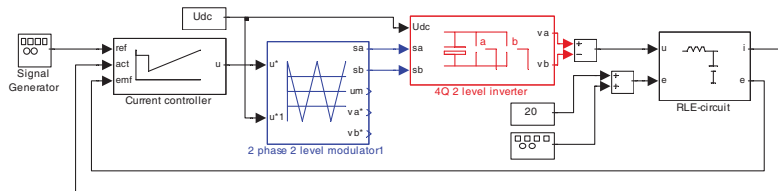


Figure 3.9: 4 quadrant DC link converter.

Assume a load circuit and a 2-quadrant DC converter with parameters according to:

$$\begin{aligned} L &= 10 \text{ mH} \\ R &= 1 \text{ } \Omega \\ T_s &= 0.5 \text{ ms} \\ U_{dc} &= 100 \text{ V} \end{aligned}$$

To demonstrate the difference to the 2 quadrant converter a simulation with the emf $e=20$ V is made. In the 2-quadrant the performance at the negative current step was rather slow (see Figure 3.8) but the 4-quadrant converter is able to supply a negative voltage, thus a faster current fall. Figure 3.10 shows the simulation results.

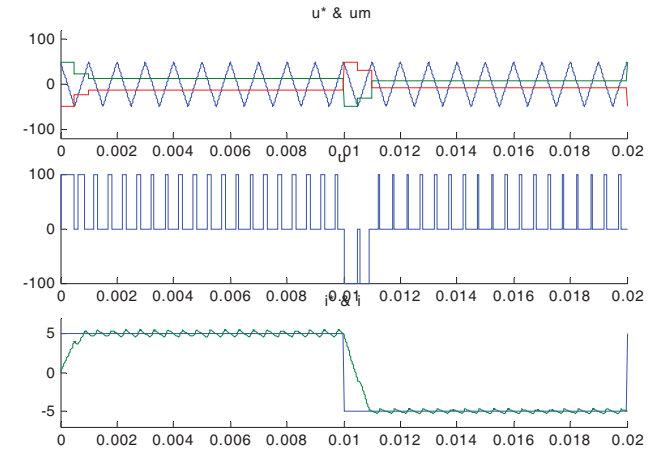


Figure 3.10: Step response with a 4-quadrant converter.

Note the fast negative current response, due to the ability of the 4-quadrant converter to supply the load with a negative load.

3.6 Direct current control of a single phase load

With a finite number of switch states it is possible to select a relevant switch state based on the instantaneous value of the current error, i.e. without using any intermediate current controller and modulator as have been used in the previous sections. This method is used in several AC motor drive types, denoted "DTC"-Direct Torque Control. It is perfectly possible to apply the same methods to DC current control with 2-quadrant or 4-quadrant converters. Some examples on how this could be done are shown in the following sections.

Direct Current Control with a 2-quadrant converter

This case is the straightest forward, since there are only two switch states to select. A switch function is applied to the current error, selecting the proper switch state according to equation (3.14). Note that the switch state is denoted $[-1,1]$ instead of $[0,1]$ as has been used before. This change is introduced due to simplicity in the implementation.

$$s = \begin{cases} 1 & \text{if } i < i^* - \frac{\Delta i}{2} \\ -1 & \text{if } i > i^* + \frac{\Delta i}{2} \\ s & \text{if } i^* - \frac{\Delta i}{2} < i < i^* + \frac{\Delta i}{2} \end{cases} \quad (3.14)$$

where Δi is the allowed current ripple in the load current. The switch function defined by equation (3.14) is easily implemented in a relay with hysteresis. Figure 3.11 shows the top-level block diagram of a Simulink model of a 2-quadrant DC converter with a Direct Current Controller according to equation (3.14).

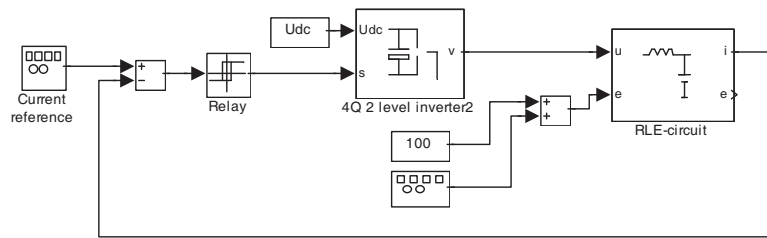


Figure 3.11: 2-quadrant DC converter with a Direct Current Controller implemented as a relay.

The switching frequency is not constant with this type of controller, since the transition time between the two extreme values of the current ripple depends on the emf e in relation to the applied voltage u . When e approaches u the transition time and thus the switching frequency approaches zero. Figure 3.12 shows an example of a simulation with the system in Figure 3.11 with an emf e that increases monotonously from $e=0$ V to $e=100$ V and a current reference being $i^*=0$ A. The other parameters are:

$$\begin{aligned} L &= 10 \text{ mH} \\ R &= 1 \text{ } \Omega \\ T_s &= 0.5 \text{ ms} \\ U_{dc} &= 100 \text{ V} \\ \Delta i &= 3 \text{ A} \end{aligned}$$

Note from Figure 3.12 that the current ripple has constant amplitude but varying frequency. The frequency variation is sometimes regarded, as a drawback since it may excite undesired resonance frequencies in nearby systems in a particular application. The audible noise from the electromagnetic forces within the reactor does have a correspondingly wide frequency content. The character of the audible noise is often put forward as a benefit with this kind of modulation, but such statements are very subjective. Figure 3.13 shows a current step with the same control system.

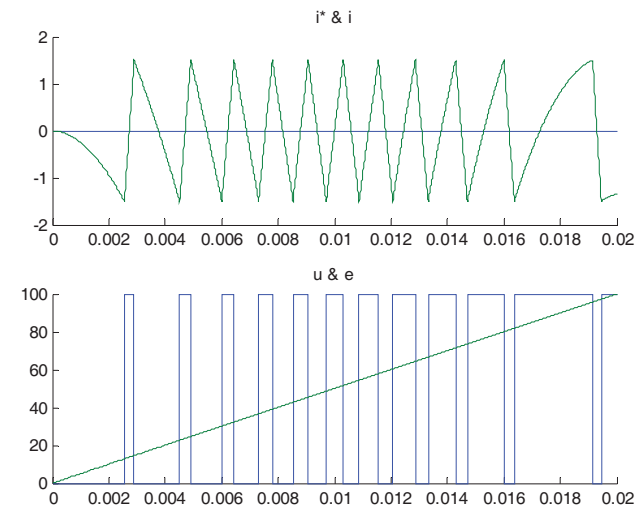


Figure 3.12: Current ripple with a varying emf e and zero current reference.

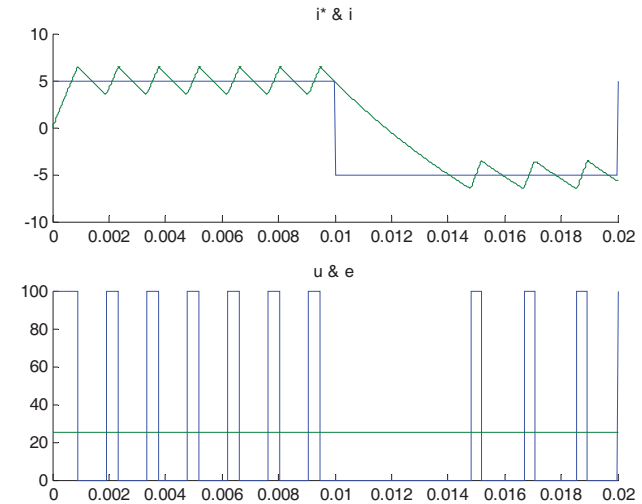


Figure 3.13: Step response with a Direct Current Controller and a 2-quadrant DC converter.

Note the fast step response, i.e. no time delay elapses between a change of the reference and a setting of new switch states. This is regarded as the major advantage with direct current

control, but it requires that the processor that implements the control operates with high sampling frequency, since it has to follow the current all the time.

Direct Current Control with a 4-quadrant converter

This case is a bit more difficult to handle than the 2-quadrant case, since there are 4 switch states to select, but only three output voltages $[U_{dc}, 0, -U_{dc}]$. One solution to this problem is to apply two tolerance bands to the current ripple, one slightly wider than the other. To understand the point with this, consider what happens when controlling a current against a positive load emf e . To increase the current, it is necessary to use the switch state $[1, -1]$, i.e. the left branch connected “up” to $U_{dc}/2$, and the right branch connected “down” to $-U_{dc}/2$. When the current thus increases it eventually hits the top of the tolerance band, and a new switch state has to be selected to force the current down again. The new switch state can be either $[-1, 1]$ or one of the alternatives $[1, 1]$ or $[-1, -1]$. The two latter ones represent a zero voltage applied to the load, and would be enough since the current will decrease under the influence of the load emf e . The active state $[-1, 1]$ will of course also work, but has the drawback that both switches are operated at the same time. Another drawback with choosing the active state $[-1, 1]$ to reduce the current is that the descent of the current will be faster, under the influence of both the emf e and the applied voltage $u = -U_{dc}$. The transition to the negative limit of the current ripple will be unnecessary fast and generate unnecessary high switching frequency. A switching strategy that involves the inactive states $[-1, -1]$ and $[1, 1]$ is thus advantageous, since it will reduce the number of switchings.

Figure 3.14 shows a tolerance band that involves the passive states as well. The tolerance band is designed with two widths. At very high current errors, a suitable active state is always chosen. When the current error decreases, after the use of an active state, and eventually hits the inner tolerance band a passive state is selected.

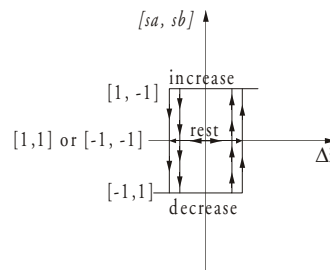


Figure 3.14: 4 quadrant DCC hysteresis including the passive states.

Assume like in the previous section that the current is to be controlled against a positive emf e , starting with a positive step higher than the tolerance bandwidth. The sequence of states is generated according to the following:

- 1 The current error is large and positive so the active state $[1, -1]$ is selected. The current increases, and the current error correspondingly decreases.

- 2 When the current error has become negative it eventually hits the inner tolerance band in the 2nd quadrant in Figure 3.14. A passive state is then selected, and the current will fall under the influence of the positive emf e , and the current error will after a while become positive.
- 3 When the current error hits the inner tolerance band level in the 1st quadrant of Figure 3.14, the positive active state $[1, -1]$ will be selected again. The current increases, and the current error correspondingly decreases. Proceed to step 2 above ...
- 4 If the current error becomes negative and higher than the inner tolerance band (e.g. due to a big negative step in the current reference), the active negative state $[-1, 1]$ is selected. The current error will after a while become positive again and hit the inner tolerance band in the 4th quadrant in Figure 3.14. A passive state is then selected. If the emf $e > 0$, as assumed in this example, then the current will continue to decrease under the influence of e and the current error will become even more positive and finally hit the outer tolerance band and set the active state $[1, -1]$. Then the current will grow and the current error will decrease again.

Since there are two passive states, a rule must be made on which of the two states that should be used at a given time. It is favorable to use both passive states approximately the same number of times since the losses that inevitable will be dissipated in the semiconductors thus will be distributed evenly in all semiconductors in the bridge. A simple rule on how to do this is thus to change the passive state every time a passive state is selected.

Figure 3.15 upper shows the top level of a Simulink simulation program with a hysteresis controller implementing the double tolerance band given in Figure 3.14, and Figure 3.15 lower part shows the content of the current controller block.

Note in Figure 3.15 that the upper hysteresis has the inner (narrower) hysteresis width and the lower hysteresis block has the outer (wider) hysteresis width. Note that the block named “Toggle” toggles between state $[1, 1]$ and $[-1, -1]$ every time a passive state is called.

In Figure 3.16 a simulation example is given with the following input data:

$$\begin{aligned} L &= 10 \text{ mH} \\ R &= 1 \text{ } \Omega \\ U_{dc} &= 100 \text{ V} \\ \Delta i &= 4 \text{ A} \end{aligned}$$

Note from Figure 3.16 that the only time the negative active state $[-1, 1]$ is used is at the negative transition of the current reference. Note also that the two passive states are altered every time a passive state is used.

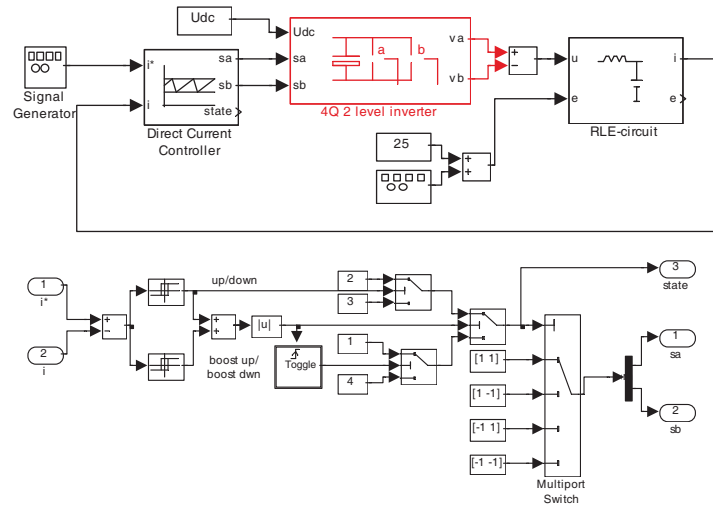


Figure 3.15: 4-quadrant DC converter with a Direct Current Controller.

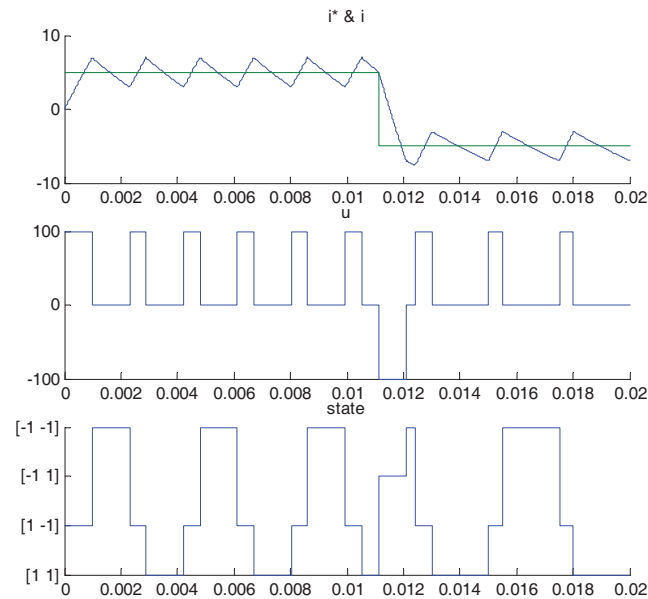


Figure 3.16: Step response with a 4 quadrant Direct Current Controller according to Figure 3.15.

3.7 Sampled current control of a three-phase converter connected to a generic three phase load

A sampled current controller for the three-phase load can be derived in the same way as with the one phase load. The main difference is that the scalar quantities used in the description of the load (3.6) and the current controller (equations (3.11) and (3.13)) is replaced with the corresponding vector quantities. The controllers derived for the 3-phase case is thus denoted vector controllers. Vector control is preferably performed in the rotating reference frame (d, q) (see Figure 3.4) instead of the stationary reference frame (α, β). The reason is that in (α, β) all signals are AC signals in stationary operation, and a PIE controller as the one derived for the scalar case (equations (3.11) and (3.13)) will give a stationary control error (eliminating the current error given at sample instant k will still leave a current error at instant $(k+1)$ since the references are continuously changing. This could theoretically be overcome with a derivating term in the controller, but derivation of the very noisy signals that are normally the case in Power Electronic Control is not feasible. In the rotating reference frame (d, q) all quantities are DC quantities in stationary operation and no stationary control error will appear.

Assume that the current is given at the sampling time instant as $\vec{i}(k)$ and the current reference at the same instant $\vec{i}^*(k)$. Furthermore, it is assumed that the current error $(\vec{i}^*(k) - \vec{i}(k))$ will be eliminated during the next sampling interval, i.e. during the interval $[k, k+1]$. The voltage needed during this interval can thus be derived from the voltage equation (3.6) by integration over one sampling interval T_s according to equation (3.15).

$$\begin{aligned} \frac{\int_{k \cdot T_s}^{(k+1)T_s} \vec{u} \cdot dt}{T_s} &= \frac{R \cdot \int_{k \cdot T_s}^{(k+1)T_s} \vec{i} \cdot dt + L \cdot \int_{k \cdot T_s}^{(k+1)T_s} \frac{d\vec{i}}{dt} \cdot dt + j \cdot \omega \cdot L \cdot \int_{k \cdot T_s}^{(k+1)T_s} \vec{i} \cdot dt}{T_s} = \quad (3.15) \\ &= \vec{u}(k, k+1) = (R + j \cdot \omega \cdot L) \cdot \vec{i}(k, k+1) + L \cdot \frac{\vec{i}(k+1) - \vec{i}(k)}{T_s} + \vec{e}(k, k+1) \end{aligned}$$

To convert the “average voltage”-equation (3.15) into a current control algorithm some assumptions have to be made:

$$\begin{aligned} \vec{u}(k, k+1) &= \vec{u}^*(k) & (a) \\ \vec{i}(k+1) &= \vec{i}^*(k) & (b) \\ \vec{i}(k, k+1) &= \frac{\vec{i}^*(k) + \vec{i}(k)}{2} & (c) \\ \vec{e}(k, k+1) &= \vec{e}(k) & (d) \\ \vec{i}(k) &= \sum_{n=0}^{n=k-1} (\vec{i}^*(n) - \vec{i}(n)) & (e) \end{aligned} \quad (3.16)$$

Assumption (a) is straight forward, the reference voltage for the coming period should be the voltage needed to achieve the desired current change. Assumption (b) is an assumption of “dead beat” current control, i.e. the entire current error is eliminated in one sampling interval. Assumption (c) is an approximation, and is not always true, since the trajectory of the current will depend on the relation between the load emf $\bar{e}$, the dc link voltage U_d and the sequence of switching states that is used within the interval $[k, (k+1)]$. However, the assumption (c) is true in average, and furthermore it is related to the resistive voltage drop that normally is relatively small, thus as small error in this assumption is not expected to distort the current control performance significantly. The assumption (d) is usually true since the sampling frequency usually is several orders of magnitude faster than the dynamics of the load emf. In the case with control of the current to the utility grid, $\bar{e}$ varies slowly or not at all when expressed in the (d, q) reference frame. The assumption (e) is a consequence of the “dead beat” assumption (b). The current at instant k is the sum of all changes of currents until the last change depicted in the previous sampling instant. With all these assumptions a current controller can be derived from the “average voltage” equation (3.9) as:

$$\begin{aligned}
 \bar{u}^*(k) &= (R + j \cdot \omega \cdot L) \cdot \frac{\bar{i}^*(k) + \bar{i}(k)}{2} + L \cdot \frac{\bar{i}^*(k) - \bar{i}(k)}{T_s} + \bar{e}(k) = \\
 &= R \cdot \frac{\bar{i}^*(k) - \bar{i}(k)}{2} + R \cdot \bar{i}(k) + L \cdot \frac{\bar{i}^*(k) - \bar{i}(k)}{T_s} + j \cdot \omega \cdot L \cdot \frac{\bar{i}^*(k) + \bar{i}(k)}{2} + \bar{e}(k) \approx \\
 &\approx \left(\frac{L}{T_s} + \frac{R}{2} \right) (\bar{i}^*(k) - \bar{i}(k)) + R \cdot \sum_{n=0}^{n=k-1} (\bar{i}^*(n) - \bar{i}(n)) + j \cdot \omega \cdot L \cdot \bar{i}(k) + \bar{e}(k) = \\
 &= \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left(\underbrace{(\bar{i}^*(k) - \bar{i}(k))}_{\text{Proportional}} + \underbrace{\left(\frac{L}{R} + \frac{T_s}{2} \right) \cdot \sum_{n=0}^{n=k-1} (\bar{i}^*(n) - \bar{i}(n))}_{\text{Integral}} \right) + \underbrace{j \cdot \omega \cdot L \cdot \bar{i}(k) + \bar{e}(k)}_{\text{Feed forward}}
 \end{aligned} \tag{3.17}$$

Note that the additional approximation introduced in equation (3.17) is the inductive voltage drop, which is relatively small compared to the proportional part of the control algorithm in transients and to the emf in stationary operation. Note also that a “PIE” current controller has been derived, with both gain and integral time constant directly related to the circuit parameters, and with a feed forward term directly related to the inductive voltage drop and the emf of the load. These properties are useful since it is now possible to determine the controller sensitivity to changes of a particular parameter, e.g. saturation dependent inductances and temperature dependent resistances.

The controller of equation (3.17) becomes more clear when split up in its two components according to equation (3.18).

$$\begin{aligned}
 u_d^*(k) &= \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left((\bar{i}_d^*(k) - \bar{i}_d(k)) + \left(\frac{L}{R} + \frac{T_s}{2} \right) \cdot \sum_{n=0}^{n=k-1} (\bar{i}_d^*(n) - \bar{i}_d(n)) \right) - \omega \cdot L \cdot \bar{i}_q(k) \\
 u_q^*(k) &= \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left((\bar{i}_q^*(k) - \bar{i}_q(k)) + \left(\frac{L}{R} + \frac{T_s}{2} \right) \cdot \sum_{n=0}^{n=k-1} (\bar{i}_q^*(n) - \bar{i}_q(n)) \right) + \omega \cdot L \cdot \bar{i}_d(k) + e_q(k)
 \end{aligned} \tag{3.18}$$

These two components are scalar and can readily be implemented as two interrelated PIE-controllers in a vector control system for the current to a generic 3-phase load.

Sampled current control of a generic 3-phase load with a slow computer

In a previous section it was discussed how to handle the problem with a delay when controlling a 1-phase current. The identical problem occurs when controlling a three phase current with the vector controller discussed in the previous section. The solution is also the same. A simplified “dummy” system model is used, with the same dynamics as the system that we are controlling. The “dummy” model current is used to calculate the change a certain voltage reference will accomplish in the controlled current, and to use this estimated change when predicting the controlled current “one sample ahead”.

$$\bar{u} = R \cdot \bar{i} + L \cdot \frac{d\bar{i}}{dt} + j \cdot \omega \cdot L \cdot \bar{i} \quad (\text{NOTE! No emf modelled})$$

$$\bar{u}^*(k) = \bar{u}(k, k+1) = R \cdot \bar{i}_{sp}(k) + j \cdot \omega \cdot L \cdot \bar{i}_{sp}(k) + \frac{L}{T_s} \cdot (\bar{i}_{sp}(k+1) - \bar{i}_{sp}(k))$$

or, in the discrete domain;

$$\begin{aligned}
 \bar{u} &= R \cdot \bar{i}_{sp} + j \cdot \omega \cdot L \cdot \bar{i}_{sp} + \frac{L}{T_s} \cdot (z-1) \cdot \bar{i}_{sp} \\
 \frac{\bar{i}_{sp}}{\bar{u}} &= \frac{1}{R + j \cdot \omega \cdot L + \frac{L}{T_s} \cdot (z-1)} = \frac{T_s}{L} \cdot \frac{1}{z - (1 - R \cdot T_s - j \cdot \omega \cdot T_s)}
 \end{aligned}$$

or, again in the time domain;

$$\bar{i}_{sp}(k+1) = \bar{i}_{sp}(k) \cdot (1 - R \cdot T_s / L - j \cdot \omega \cdot T_s) + \bar{u}^*(k) \cdot \frac{T_s}{L}$$

The current estimate to be used by the controller of the “slow computer” is thus finally calculated as:

$$\hat{\bar{i}}(k+1) = \bar{i}(k) + (\bar{i}_{sp}(k+1) - \bar{i}_{sp}(k)) \tag{3.19}$$

The accuracy of the current prediction is of course dependent on the accuracy of the parameters and the estimate of the induced voltage used by the controller. A simple sensitivity analysis is presented in the next section. The current controller algorithm when using a slow computer is thus given by equation (3.20).

$$u_d^*(k) = \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left(\hat{i}_d^*(k) - \hat{i}_d(k) \right) + \frac{T_s}{\left(\frac{L}{R} + \frac{T_s}{2} \right)} \cdot \sum_{n=0}^{n=k-1} (\hat{i}_d^*(n) - \hat{i}_d(n)) - \omega \cdot L \cdot \hat{i}_q(k) \quad (3.20)$$

$$u_q^*(k) = \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot \left(\hat{i}_q^*(k) - \hat{i}_q(k) \right) + \frac{T_s}{\left(\frac{L}{R} + \frac{T_s}{2} \right)} \cdot \sum_{n=0}^{n=k-1} (\hat{i}_q^*(n) - \hat{i}_q(n)) + \omega \cdot L \cdot \hat{i}_d(k) + e_q(k)$$

The quality of the current prediction depends on application. With a 3-phase load like the one used in this section, it is possible to measure e.g. the emf $\vec{e}$, and the inductor parameters are usually rather easy to measure. When controlling e.g. a AC machine, the emf is not available for measurement and the estimate is of course more difficult to make. It remains to be determined in the specific application what methods can be used, and with what accuracy, to estimate circuit parameters and the emf.

Sampled Vector Current control applied to a 3-phase load with a 2 level converter

The vector current controller derived in the previous chapter will now be applied to a 2 level 3-phase VSI connected to a 3-phase generic load. Figure 3.17 shows a Simulink block diagram of the system under consideration.

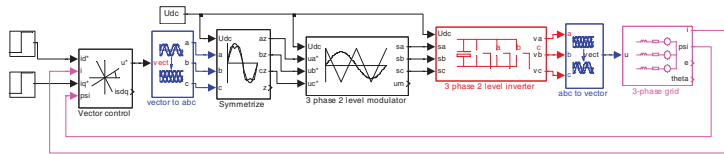


Figure 3.17: A 3-phase converter with vector control connected to a generic 3-phase load

The blocks visible in Figure 3.17 contain the following functions:

Vector Control: A vector controller according to equation (3.18) or equation (3.20), selectable.

Vector to abc: 2-phase to 3-phase transformation, power invariant.

3 phase 2 level modulator and Inverter: Carrier wave modulator and 3 ideal switches according to chapter 3.

Abc to vector: 3-phase to 2-phase conversion, power invariant.

3-phase grid: Ideal 3 phase load.

The contents of the current control block are shown in Figure 3.18.

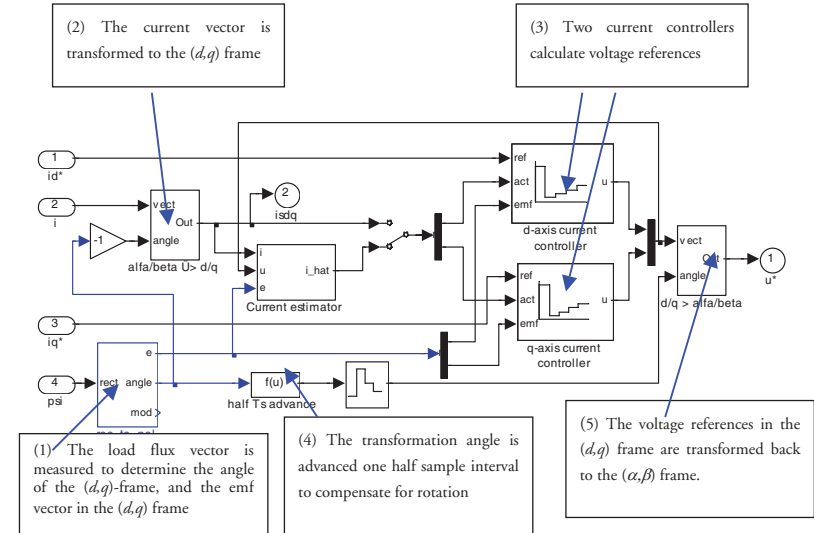


Figure 3.18: Content of the current control block.

Assume a load circuit and a 3-phase converter with parameters according to:

$$L = 10 \text{ mH}$$

$$R = 1 \text{ } \Omega$$

$$T_s = 0.5 \text{ ms}$$

$$U_{dc} = 750 \text{ V}$$

To evaluate the performance of the developed current controllers, some simulations are made. In all cases both the d -axis and the q -axis currents are making step changes between ± 10 A. The following situations are simulated:

- I. The current controller has the right set of parameters, i.e. a perfect measurement of the load parameters and an as perfect measurement of the load emf is used. The controller is of the "fast computer" type.

- II. In this case the current controller is of the slow computer type, and the controller parameters are deliberately different from those of the load, to show the performance of a mistuned current controller. One case is simulated; with $R_{\text{controller}}=1.5R$ and $L_{\text{controller}}=0.5L$. This is an extreme mistuning of the controller. Normally the parameters are known with less than 10 % error, but the mistuning is chosen to show the abilities of the current estimator.

Figure 3.19 shows some simulation results from case I. Note that the phase current increases in amplitude at the q -axis current step, and that the phase is advanced as well. This is due to the increase amplitude, the modulus of the vector sum of i_d and i_q , and the fact that the current vector advances 45 degrees as the q -axis current increases.

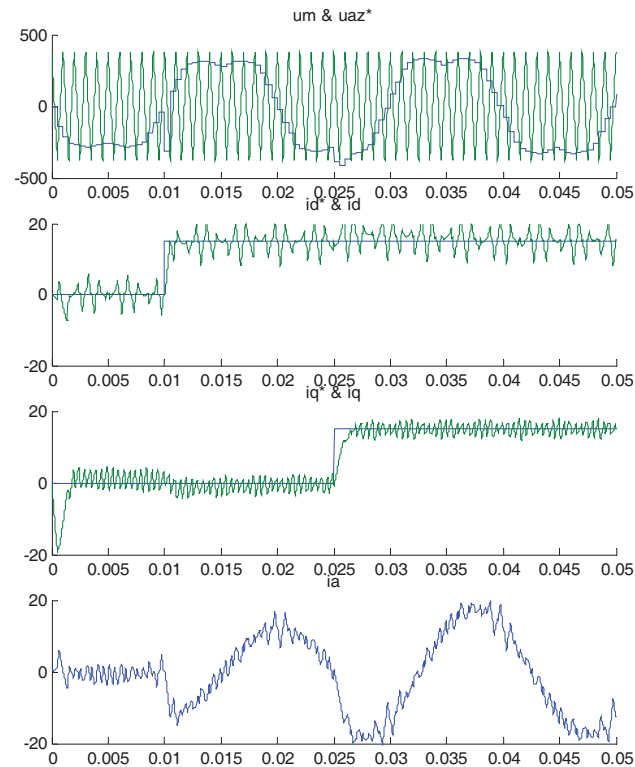


Figure 3.19: Vector control of 3-phase converter connected to a generic 3-phase load with a fast computer.

Figure 3.20 shows the result of case II above. Note that no significant difference in performance can be noted, in spite of the drastic mistuning of the controller parameters.

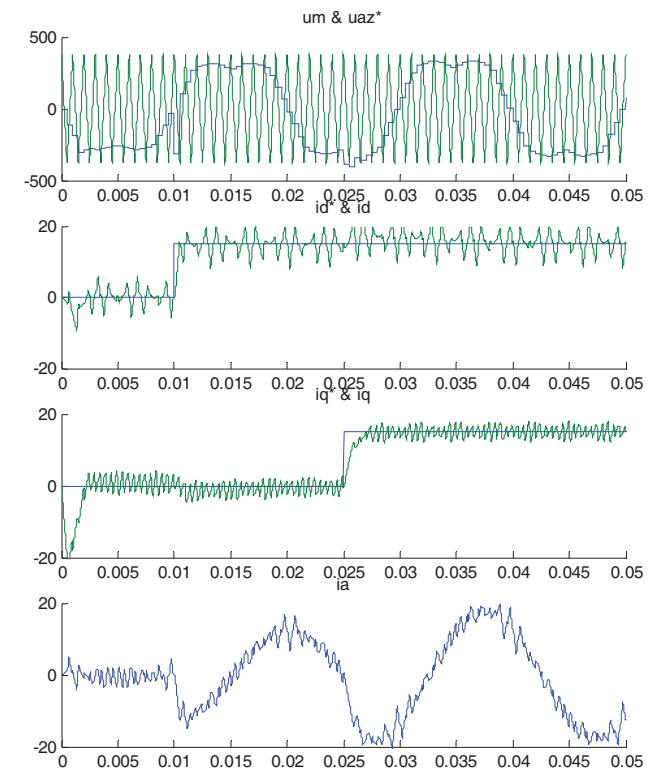


Figure 3.20: Vector control of 3-phase converter connected to a generic 3-phase load with a fast computer using current estimation.

The vector control system presented in this section is very similar to vector control systems used in ac motor control systems. The current control as such is not difficult to engineer. The biggest challenge is instead in the estimation of the load flux. In the generic load used in this section, the load flux is measurable, but in electrical machines it is generally more difficult to obtain a good estimate of the flux. This challenge will be addressed in later chapters on ac machine control.

3.8 Relation to traditional control theory

A model-based controller is derived for a grid connected voltage source converter (Figure 2.18). The controller is model-based in the sense that the controller parameters are dependent on the grid parameters, i.e. L (L_{line}) and R (R_{line}). The current controller for three-phase converters discussed here is based on the synchronously rotating (dq) reference frame.

Synchronously rotating refers to that the reference frame is rotating with the same speed as the grid voltage vector $\bar{e}^{\alpha\beta}$. The rotating reference system has two axes denoted d and q . The grid voltage vector $\bar{e}^{\alpha\beta}$ coincides with the q -axis in this case, which means that $e_d = 0$ (Figure 2.25). The transformation expressions are written

$$\bar{s}^{dq} = \bar{s}^{\alpha\beta} \cdot e^{-j\omega_1 t} \quad \text{and} \quad \bar{s}^{\alpha\beta} = \bar{s}^{dq} \cdot e^{j\omega_1 t} \quad (3.21)$$

The controller is based on a model of the grid together with the converter output filter

$$\bar{u}^{\alpha\beta} - L \frac{d}{dt} \bar{i}^{\alpha\beta} - R \bar{i}^{\alpha\beta} - \bar{e}^{\alpha\beta} = 0 \quad (3.22)$$

This is transformed to dq -coordinates by use of the transformation above, i.e.

$$\bar{u}^{dq} \cdot e^{j\omega_1 t} - L \frac{d}{dt} (\bar{i}^{dq} \cdot e^{j\omega_1 t}) - R \bar{i}^{dq} \cdot e^{j\omega_1 t} - \bar{e}^{dq} \cdot e^{j\omega_1 t} = 0 \quad (3.23)$$

which gives

$$\bar{u}^{dq} \cdot e^{j\omega_1 t} - L \frac{d}{dt} \bar{i}^{dq} \cdot e^{j\omega_1 t} - j\omega_1 L \bar{i}^{dq} \cdot e^{j\omega_1 t} - R \bar{i}^{dq} \cdot e^{j\omega_1 t} - \bar{e}^{dq} \cdot e^{j\omega_1 t} = 0 \quad (3.24)$$

Simplifying the last expression gives

$$\bar{u}^{dq} - L \frac{d}{dt} \bar{i}^{dq} - j\omega_1 L \bar{i}^{dq} - R \bar{i}^{dq} - \bar{e}^{dq} = 0 \quad (3.25)$$

By dividing into its components, i.e. real and imaginary parts, the resulting expressions are

$$\begin{cases} u_d = L \frac{d}{dt} i_d + R i_d - \omega_1 L i_q + e_d \\ u_q = L \frac{d}{dt} i_q + R i_q + \omega_1 L i_d + e_q \end{cases} \quad (3.26)$$

Controller design based on understanding

A discrete-time, dead-beat PI-controller is designed for each component based on the last two expressions. The output from the controller is two voltage references, one in d - and one in q -direction. The first step is to use backward Euler approximation [25] for the current derivatives. This gives

$$\begin{cases} \frac{di_d}{dt} \approx \frac{i_{d,k} - i_{d,k-1}}{T_s} \\ \frac{di_q}{dt} \approx \frac{i_{q,k} - i_{q,k-1}}{T_s} \end{cases} \quad (3.27)$$

Furthermore, it is assumed that the currents can be approximated with their average values, in the particular sampling interval

$$\begin{cases} i_d \approx \frac{i_{d,k} + i_{d,k-1}}{2} \\ i_q \approx \frac{i_{q,k} + i_{q,k-1}}{2} \end{cases} \quad (3.28)$$

It is also assumed that the grid voltage does not change between two samples. Consequently,

$$\begin{cases} e_{d,k} \approx e_{d,k-1} \\ e_{q,k} \approx e_{q,k-1} \end{cases} \quad (3.29)$$

This gives the following result

$$\begin{cases} u_{d,k} = L \frac{i_{d,k} - i_{d,k-1}}{T_s} + R \frac{i_{d,k} + i_{d,k-1}}{2} - \omega_1 L \frac{i_{q,k} + i_{q,k-1}}{2} + e_{d,k-1} \\ u_{q,k} = L \frac{i_{q,k} - i_{q,k-1}}{T_s} + R \frac{i_{q,k} + i_{q,k-1}}{2} + \omega_1 L \frac{i_{d,k} + i_{d,k-1}}{2} + e_{q,k-1} \end{cases} \quad (3.30)$$

Since a dead-beat controller is considered, it is assumed that

$$\begin{cases} i_{d,k} = i_{d,ref,k-1} \\ i_{q,k} = i_{q,ref,k-1} \end{cases} \quad (3.31)$$

giving

$$\begin{cases} u_{d,k} = L \frac{i_{d,ref,k} - i_{d,k}}{T_s} + R \frac{i_{d,ref,k} + i_{d,k}}{2} - \omega_1 L \frac{i_{q,ref,k} + i_{q,k}}{2} + e_{d,k} \\ u_{q,k} = L \frac{i_{q,ref,k} - i_{q,k}}{T_s} + R \frac{i_{q,ref,k} + i_{q,k}}{2} + \omega_1 L \frac{i_{d,ref,k} + i_{d,k}}{2} + e_{q,k} \end{cases} \quad (3.32)$$

This gives a P-controller according to

$$\begin{cases} u_{d,ref,k} = \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot (i_{d,ref,k} - i_{d,k}) + R \cdot i_{d,k} - \omega_1 L \frac{i_{q,ref,k} + i_{q,k}}{2} + e_{d,k} \\ u_{q,ref,k} = \left(\frac{L}{T_s} + \frac{R}{2} \right) \cdot (i_{q,ref,k} - i_{q,k}) + R \cdot i_{q,k} + \omega_1 L \frac{i_{d,ref,k} + i_{d,k}}{2} + e_{q,k} \end{cases} \quad (3.33)$$

The resistive voltage drop term can be interpreted as an integral part assuming that the current equals the sum of all the previous current errors, i.e.

$$\begin{cases} i_{d,k} = \sum_{n=0}^{k-1} (i_{d,ref,n} - i_{d,n}) \\ i_{q,n} = \sum_{n=0}^{k-1} (i_{q,ref,n} - i_{q,n}) \end{cases} \quad (3.34)$$

This gives a PI-controller

$$\begin{cases} u_{d,ref,k} = K \cdot \left((i_{d,ref,k} - i_{d,k}) + \frac{1}{T_i} \cdot \sum_{n=0}^{k-1} (i_{d,ref,n} - i_{d,n}) \right) - K_c \cdot \frac{i_{q,ref,k} + i_{q,k}}{2} + e_{d,k} \\ u_{q,ref,k} = K \cdot \left((i_{q,ref,k} - i_{q,k}) + \frac{1}{T_i} \cdot \sum_{n=0}^{k-1} (i_{q,ref,n} - i_{q,n}) \right) + K_c \cdot \frac{i_{d,ref,k} + i_{d,k}}{2} + e_{q,k} \end{cases} \quad (3.35)$$

where

$$\begin{cases} K = \left(\frac{L}{T_s} + \frac{R}{2} \right) \\ T_i = R / \left(\frac{L}{T_s} + \frac{R}{2} \right) = 1 / \left(\frac{L}{RT_s} + \frac{1}{2} \right) \\ K_c = \frac{\omega_1 L}{2} \end{cases} \quad (3.36)$$

RST controller design

This design of the same controller as determined previously follows the method described in Chapter 10 of [25]. The plant is the grid including line filter expressed in the rotating dq -frame, which is described by the first order differential equation

$$\frac{d}{dt} \vec{i}^{dq} = - \left(\frac{R}{L} + j\omega_1 \right) \vec{i}^{dq} + \frac{1}{L} \vec{u}^{dq} - \frac{1}{L} \vec{e}^{dq} \quad (3.37)$$

Each component is described by

$$\begin{cases} \frac{di_d}{dt} = -\frac{R}{L} i_d + \omega_1 i_q + \frac{1}{L} u_d - \frac{1}{L} e_d \\ \frac{di_q}{dt} = -\frac{R}{L} i_q - \omega_1 i_d + \frac{1}{L} u_q - \frac{1}{L} e_q \end{cases} \quad (3.38)$$

The voltage $\vec{u}^{dq} = u_d + j u_q$ is the output of the controller and the current $\vec{i}^{dq} = i_d + j i_q$ is the controlled quantity. Therefore the term containing the grid voltage $\vec{e}^{dq} = e_d + j e_q$ and the cross-coupling terms $j\omega_1 \vec{i}^{dq} = -i_q + j i_d$ can be regarded as known disturbances that are compensated by means of feed forward. This implies that the plant can be expressed as

$$\frac{d}{dt} \vec{i}^{dq} = -\frac{R}{L} \vec{i}^{dq} + \frac{1}{L} \vec{u}^{dq} + \vec{w}^{dq} \quad (3.39)$$

where the disturbance is

$$\vec{w}^{dq} = -j\omega_1 \vec{i}^{dq} - \frac{1}{L} \vec{e}^{dq} \quad (3.40)$$

The transfer function from voltage $\vec{u}^{dq}$ to current $\vec{i}^{dq}$ of the plant is thus written

$$G_{plant}(s) = \frac{B(s)}{A(s)} = \frac{1/L}{s + R/L} = (1/R) \frac{R/L}{s + R/L} \quad (3.41)$$

The next step is to discretize the plant equation. There are several methods to accomplish this for example zero-order-hold (zoh) sampling or forward Euler approximation. In case of zoh sampling the transfer function is

$$G_{plant}(q) = \frac{B(q)}{A(q)} = \frac{(1/R)(1 - e^{-RT_s/L})}{q - e^{-RT_s/L}} \approx \frac{(1/R)(RT_s/L)}{q - (1 - RT_s/L)} = \frac{T_s/L}{q - (1 - RT_s/L)} \quad (3.42)$$

where the approximated transfer function above is the same as the one obtained with forward Euler approximation i.e.

$$\vec{i}_{k+1}^{dq} = \left(1 - \frac{RT_s}{L} \right) \vec{i}_k^{dq} + \frac{T_s}{L} \vec{u}_k^{dq} \quad (3.43)$$

Note that the converter operates in switch-mode, which means that the current can change substantially between sampling instants in the case of a change of reference current. Therefore, the resistive voltage drop is sometimes written as the average current for that particular sampling interval i.e.

$$R \vec{i}_{[k,k+1]}^{dq} \approx R \frac{\vec{i}_{k+1}^{dq} + \vec{i}_k^{dq}}{2} = \frac{R}{2} (\vec{i}_{k+1}^{dq} + \vec{i}_k^{dq}) \quad (3.44)$$

Applying this to the forward Euler approximated transfer function gives

$$\vec{i}_{k+1}^{dq} = \frac{2L - RT_s}{2L + RT_s} \vec{i}_k^{dq} + \frac{2T_s}{2L + RT_s} \vec{u}_k^{dq} \quad (3.45)$$

In this case the transfer function is written

$$G_{plant}(q) = \frac{B(q)}{A(q)} = \frac{\frac{2T_s}{2L+RT_s}}{q - \frac{2L-RT_s}{2L+RT_s}} \quad (3.46)$$

An RST controller has the form

$$\vec{u}_k^{dq,ref} = \frac{T(q)}{R(q)} \vec{i}_k^{dq,ref} - \frac{S(q)}{R(q)} \vec{i}_k^{dq} \quad (3.47)$$

If an RST controller is employed, the resulting closed loop system is

$$\vec{i}_k^{dq} = \frac{B(q)}{A(q)} \left[\frac{T(q)}{R(q)} \vec{i}_k^{dq,ref} - \frac{S(q)}{R(q)} \vec{i}_k^{dq} \right] \quad (3.48)$$

Rearranging the terms yields

$$[A(q)T(q) + B(q)S(q)] \vec{i}_k^{dq} = B(q)T(q) \vec{i}_k^{dq,ref} \quad (3.49)$$

In this case a PI controller (with feed-forward) is utilized. A PI controller is written

$$\vec{u}_k^{dq,ref} = K_i \left(\vec{i}_k^{dq,ref} - \vec{i}_k^{dq} \right) + \Sigma \Delta \vec{i}_k^{dq} \quad (3.50)$$

where $\Sigma \Delta \vec{i}_k^{dq}$ is the integral part which is updated according to

$$\Sigma \Delta \vec{i}_{k+1}^{dq} = \Sigma \Delta \vec{i}_k^{dq} + \frac{K_i}{T_i} \left(\vec{i}_k^{dq,ref} - \vec{i}_k^{dq} \right) \quad (3.51)$$

Note that integral part anti wind-up should be implemented based on back-calculation of the maximum permissible voltage vector since over-modulation may occur. The transfer function of the PI controller is thus

$$\vec{u}_k^{dq,ref} = K_i \left(1 + \frac{1/T_i}{q-1} \right) \left(\vec{i}_k^{dq,ref} - \vec{i}_k^{dq} \right) = \frac{K_i q + K_i (1/T_i - 1)}{q-1} \left(\vec{i}_k^{dq,ref} - \vec{i}_k^{dq} \right) \quad (3.52)$$

This means that

$$\begin{cases} R(q) = q-1 \\ S(q) = T(q) = K_i q + K_i (1/T_i - 1) \end{cases} \quad (3.53)$$

According to [25] $R(q)$ should be monic which it is. For causality the order of $R(q)$ should be higher or equal to the order of $S(q)$ and $T(q)$ which is also the case here. Since $A(q)$, $R(q)$ and $S(q)$ is of the first order and $B(q)$ is of zero order it follows that the closed loop transfer

function has at most two poles (if no poles are cancelled). Since $S(q) = T(q)$, the poles introduced by $S(q)$ can be cancelled if $S(q)$ is a factor of $A(q)$. Assume that

$$\begin{cases} A(q) = q + a_0 \\ B(q) = b_0 \end{cases} \quad (3.54)$$

This means that (3.49) rewritten as

$$\begin{aligned} [(q+a_0)(q-1) + b_0(K_i q + K_i(1/T_i - 1))] \vec{i}_k^{dq} &= \\ = b_0(K_i q + K_i(1/T_i - 1)) \vec{i}_k^{dq,ref} \end{aligned} \quad (3.55)$$

Rearranging gives

$$\begin{aligned} [q^2 + (a_0 - 1 + b_0 K_i)q + (b_0 K_i(1/T_i - 1) - a_0)] \vec{i}_k^{dq} &= \\ = [b_0 K_i q + b_0 K_i(1/T_i - 1)] \vec{i}_k^{dq,ref} \end{aligned} \quad (3.56)$$

Dead-beat control implies that the actual current should reach the reference in a time equal to the sampling interval times the order of the plant. In this case the order of the plant is one and therefore the denominator of the closed loop system should be equal to one. Furthermore, for dead-beat control the denominator should have its poles at $q=0$. This means that the denominator of the closed loop transfer function should be $A_m(q) = q$. Therefore, the lowest order term of the left hand expression above should be equal to zero and that $S(q) = T(q)$ should be a factor of $A(q)$ i.e.

$$\begin{cases} b_0 K_i (1/T_i - 1) - a_0 = 0 \\ a_0 = 1/T_i - 1 \end{cases} \quad (3.57)$$

which gives

$$\begin{cases} K_i = 1/b_0 \\ T_i = 1/(a_0 + 1) \end{cases} \quad (3.58)$$

Consequently

$$\begin{cases} a_0 = (RT_s/L) - 1 \\ b_0 = T_s/L \end{cases} \Rightarrow \begin{cases} K_i = L/T_s \\ T_i = L/RT_s \end{cases} \quad (3.59)$$

and

$$\begin{cases} a_0 = \frac{RT_s - 2L - 2L + RT_s}{2L + RT_s} \\ b_0 = \frac{2T_s}{2L + RT_s} \end{cases} \Rightarrow \begin{cases} K_i = \frac{L}{T_s} + \frac{R}{2} \\ T_i = \frac{L}{RT_s} + \frac{1}{2} \end{cases} \quad (3.60)$$

where the latter controller parameters are identical to the ones derived in the previous section. Note that in most modern applications $L/T_s \gg R/2$ and therefore the two controller parameter sets give equal behavior.

Smith-predictor

Also in the presence of plant delay the controller parameters are derived the same way. In this case the voltage reference calculated by the current controller is delayed one sample interval due to the calculation time of the DSP, which implies that the actual transfer function of the plant is not $G_p(q)$ but $q^{-1} \cdot G_p(q)$. Consequently, after selection of appropriate controller parameters a Smith-predictor is designed to predict the current based on the applied output voltage and the previous predictions, i.e. the state of the Smith-predictor. According to [25] the the current estimated by the Smith-predictor is given by

$$\bar{i}_{Smith,k}^{dq} = (-1 + q^{-1}) G_p(q) \bar{u}_k^{dq,ref} \quad (3.61)$$

Shifting both sides one sample forward gives

$$\bar{i}_{Smith,k+1}^{dq} = (-q + 1) G_p(q) \bar{u}_k^{dq,ref} \quad (3.62)$$

Note that in this case the plant should be the full system since

$$\bar{i}_k = G_p(q) (\bar{u}_k^* - \bar{e}_k) = \frac{T_s/L}{q - (1 - T_s R/L - j\omega T_s)} (\bar{u}_k^* - \bar{e}_k) \quad (3.63)$$

which means that the Smith-predictor is

$$\begin{aligned} \bar{i}_{Smith,k+1}^{dq} &= (-q + 1) \frac{T_s/L}{q - (1 - T_s R/L - j\omega T_s)} (\bar{u}_k^* - \bar{e}_k) = \\ &= (-q + 1) \frac{T_s/L}{q - (1 - T_s R/L - j\omega T_s)} \cdot \bar{u}_k^* + \\ &+ \frac{T_s/L}{q - (1 - T_s R/L - j\omega T_s)} \cdot (-\bar{e}_{k+1} + \bar{e}_k) \approx \\ &\approx (-q + 1) \frac{T_s/L}{q - (1 - T_s R/L - j\omega T_s)} \cdot \bar{u}_k^* \end{aligned} \quad (3.64)$$

Consequently, the feed forward voltage is not taken into account (if its bandwidth is much less than the sampling frequency) but the current cross-coupling terms are taken into account.

For implementation in a DSP a new state is introduced in such a way that the Smith-predictor is written

$$\begin{cases} \bar{i}_{Smith,k+1} = \left(\frac{RT_s}{L} s_k + j\omega T_s \right) \bar{s}_k - \frac{T_s}{L} \bar{u}_k^* \\ \bar{s}_{k+1} = \left(1 - \frac{RT_s}{L} - j\omega T_s \right) \bar{s}_k + \frac{T_s}{L} \bar{u}_k^* \end{cases} \quad (3.65)$$

If this is divided between components the following is obtained

$$\begin{cases} \bar{i}_{Smith,d,k+1} = \frac{RT_s}{L} s_{d,k} - \omega T_s s_{q,k} - \frac{T_s}{L} u_{d,k}^* \\ \bar{i}_{Smith,q,k+1} = \frac{RT_s}{L} s_{q,k} + \omega T_s s_{d,k} - \frac{T_s}{L} u_{q,k}^* \end{cases} \quad (3.66)$$

$$\begin{cases} s_{d,k+1} = \left(1 - \frac{RT_s}{L} \right) s_{d,k} + \omega T_s s_{q,k} + \frac{T_s}{L} u_{d,k}^* \\ s_{q,k+1} = \left(1 - \frac{RT_s}{L} \right) s_{q,k} - \omega T_s s_{d,k} + \frac{T_s}{L} u_{q,k}^* \end{cases} \quad (3.67)$$

The current references above are added to the current references based on desired output power. The resulting voltage references, in dq -coordinates, are transformed to the fixed $\alpha\beta$ -frame. Then the $\alpha\beta$ -references are transformed to three-phase reference voltages. If the modulation is based on sinusoidal references the three-phase voltage references are used as they are. Otherwise, they are manipulated, for example to become symmetrical or bus-clamped, and then passed on to the modulator.

State-space model and dynamic properties

State-feedback according to Figure 3.21 is used in the implementation of the controller. The controller parameters in Figure 3.21 are set to $K_i = k_i ((L/T_s) + (R/2))$, $K_i/T_i = k_i R$, $K_{ci} = K_{ci}^* = \omega_1 L/2$ and $K_r = 0$. Note that $k_i = 1$ corresponds to dead-beat response. The parameters of the Smith-predictor are set to $K_{ss} = RT_s/L$, $K_{so} = \omega_1 T_s$ and $K_{su} = T_s/L$.

The output of the controller (i.e. the converter output voltage reference where * denotes reference value)

$$\bar{u}_k^* = K_i \left((\bar{i}_k^* - \bar{i}_k) + \sum \Delta \bar{i}_k \right) + K_r \bar{i}_k + jK_{ci} \bar{i}_k + jK_{ci}^* \bar{i}_k^* + \bar{e}_k \quad (3.68)$$

and the integral part

$$\sum \Delta \bar{i}_{k+1} = \sum \Delta \bar{i}_k + \frac{K_i}{T_i} (\bar{i}_k^* - \bar{i}_k) \quad (3.69)$$

$$\begin{pmatrix} i_{d,k+1} \\ i_{q,k+1} \\ u_{d,k+1} \\ u_{q,k+1} \\ s_{d,k+1} \\ s_{q,k+1} \\ \Sigma \Delta i_{d,k+1} \\ \Sigma \Delta i_{q,k+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \Phi_L & \Gamma_{Lu} & 0 & 0 \\ L_i & L_u & L_{sp} & L_{\Sigma \Delta i} \\ 0 & S_u & S_{sp} & 0 \\ I_i & I_u & I_{sp} & I_{\Sigma \Delta i} \end{pmatrix}}_{\Phi_c} \begin{pmatrix} i_{d,k} \\ i_{q,k} \\ u_{d,k} \\ u_{q,k} \\ s_{d,k} \\ s_{q,k} \\ \Sigma \Delta i_{d,k} \\ \Sigma \Delta i_{q,k} \end{pmatrix} + \underbrace{\begin{pmatrix} \Gamma_{Le} & 0 \\ L_e & L_{i^*,ext} \\ 0 & 0 \\ 0 & I_{i^*,ext} \end{pmatrix}}_{\Gamma_c} \begin{pmatrix} e_{d,k} \\ e_{q,k} \\ i_{d,ext,k}^* \\ i_{q,ext,k}^* \end{pmatrix} \quad (3.85)$$

If i_q is considered as the output, and $i_{d,ext}^*$ and $i_{q,ext}^*$ as inputs the output is described by

$$i_{q,k} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \end{pmatrix}}_{C_{c,iq}} \begin{pmatrix} i_{d,k} \\ i_{q,k} \\ u_{d,k} \\ u_{q,k} \\ s_{d,k} \\ s_{q,k} \\ \Sigma \Delta i_{d,k} \\ \Sigma \Delta i_{q,k} \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}}_{D_c} \begin{pmatrix} e_{d,k} \\ e_{q,k} \\ i_{d,ext,k}^* \\ i_{q,ext,k}^* \end{pmatrix} \quad (3.86)$$

The frequency response from active line current reference $i_{q,ext}^*$ to active line current i_q is shown in Figure 3.22 and the frequency response from reactive line current reference $i_{d,ext}^*$ to active line current i_q is shown in Figure 3.23. The parameters used to obtain the frequency responses in Figure 3.22 and Figure 3.23 are $L=11.0$ mH, $R=0.75$ Ω and $T_i=170$ μ s.

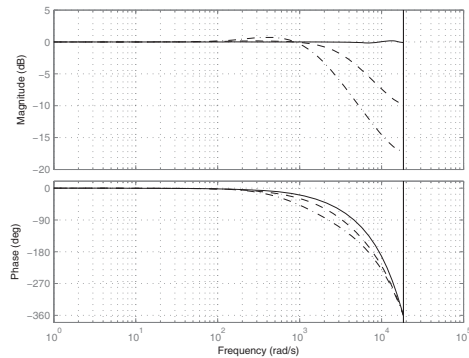


Figure 3.22 Frequency response from $i_{q,ext}^*$ to i_q for $k_i=1.0$ (solid), $k_i=0.5$ (dashed) and $k_i=0.25$ (dash-dotted).

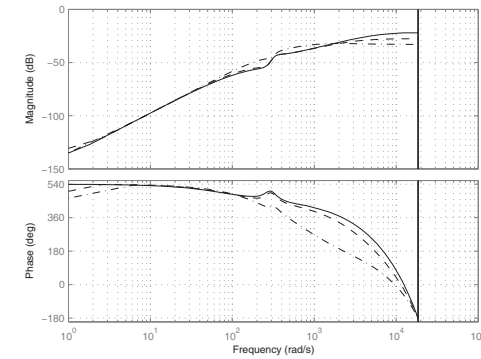


Figure 3.23: Frequency response from $i_{d,ext}^*$ to i_q for $k_i=1.0$ (solid), $k_i=0.5$ (dashed) and $k_i=0.25$ (dash-dotted).

3.9 Direct Current Control (DCC) with a 3-phase converter and a generic load

The DCC methods used in previous sections together with a 1-phase load and 2- or 4-quadrant converter can be used even with the ac converter. The selection of voltage vectors as a function of the current errors must be related to the actual position of the grid flux that defines the (d,q) -axes. The derivative of the current vector in the (α,β) frame is given by equation (3.87).

$$\frac{di}{dt} = \frac{\vec{u} - R \cdot \vec{i} - \vec{e}}{L} \quad (3.87)$$

The choice of vector must be based on the goal to force the current vector to move in a direction that reduces the vector error of the current. Figure 3.24 shows the voltage vectors that the converter is able to create, together with the (d,q) frame. See section 3.3 and Figure 3.4 for reference to the voltage vectors and the definition of the (d,q) frame. Consider the position of the (d,q) frame as shown in Figure 3.1. All vectors affect the current according to equation (3.87). To increase the d -axis current, either voltage vector 1, 2 or 6 can be used. To decrease the d -axis current, either of voltage vectors 3, 4 or 5 can be used. To increase the q -axis current the voltage vectors 2 or 3 can be used, and to reduce the q -axis current the voltage vectors 5 or 6 can be used. In addition, the zero-vectors can be used, but the effect of using them is different in the d - and q -direction. In the d -direction, there is no emf and a zero vector will have very little effect on the current derivative. In the q -direction the emf will give a significant negative derivative with a zero vector applied. Thus, the d -axis current control can be of a simpler kind that only uses active vectors in either direction to control the current, whereas the q -axis current controller benefit from using zero vectors and thus must be more advanced. The tolerance band used in the d -axis is thus the same as for the DC current with the 2-quadrant converter, see Figure 3.25 left.

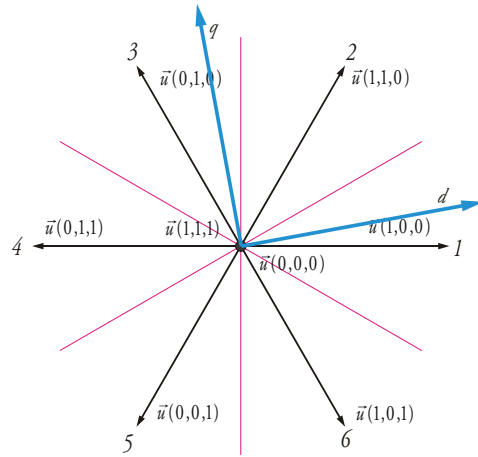


Figure 3.24: The 3-phase converter voltage vectors and the (d,q) reference frame.

The tolerance band used in the q -axis is the same as used for the DC current with the 4-quadrant converter, see Figure 3.25 right.

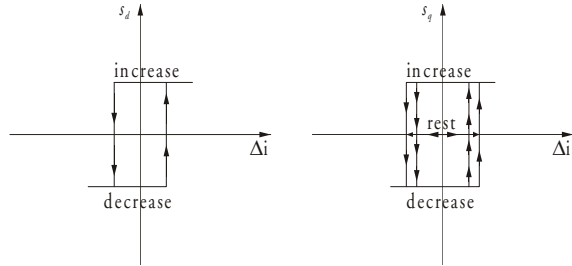


Figure 3.25: The tolerance bands used in the 3 phase DCC.

Note that the entity “rest” in Figure 3.25 implies the use of a zero vector. To proceed in defining the DCC some constraints must be introduced:

- 1 To control the q -axis current, only the active vectors closest to the q -axis must be involved. In Figure 3.24 this means vectors 2, 3, 5 and 6, and of course the zero-vectors. The other two active vectors are thus excluded from the control strategy developed here.
- 2 To identify the possible vectors to be used it is necessary to define a set of sectors surrounding the voltage vectors with $\pm \pi/6$ radians. These sectors are indicated in Figure 3.24 and given a number from 1 to 6. Sector 1 covers the angles $[-\pi/6 \dots \pi/6]$, sector 2 covers $[\pi/6 \dots \pi/2]$ etc.

With these assumptions a rule can be created on which active vectors to select based on the current errors. Table 3.1 gives the offset to the sector where the d -axis is located that should be added to the sector number to locate the correct active vector.

Table 3.1: Offsets to locate the correct voltage vector when the sector of the flux vector is known.

s_{offset}	Decrease iq	Increase iq
Decrease id	4	2
Increase id	5	1

The correct active vector can thus be calculated as

$$vector = sector + s_{offset} \quad (3.88)$$

When the q -axis current controller calls for a passive vector, the choice of active vectors is overruled. Figure 3.26 shows a Simulink program with a vector controller based on the above reasoning.

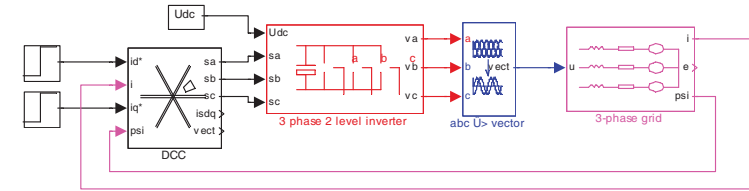


Figure 3.26: 3-phase DCC applied to a generic load.

Figure 3.27 shows the content of the DCC block in Figure 3.26. With the same motivation as in the 4-quadrant DCC it is necessary to select the zero vector in a way that minimizes and distributes the losses of the semiconductor evenly. In the 3-phase case the zero vector to be used is determined by the switch state prior to the selection of a zero vector. For any given active vector, flipping one switch can set a passive vector. E.g. starting from state $[1, -1, -1]$, the “closest” zero vector is $[-1, -1, -1]$ since only the switch in phase a has to be flipped. Thus in Figure 3.27 the selection of a suitable zero state is based on the last preceding active state.

Figure 3.28 shows the performance of the 3-phase DCC given the same conditions as for the simulations with the vector controller in section 3.7. Note how the two passive states (7 and 8) are both used based on the previous state. The DCC does appear to be ideally parameter insensitive, since the controller itself contains no load parameters at all. This is deceiving since the most difficult challenge usually is to estimate the quantity that defines the rotating reference frame, in this case the load flux. With the generic three phase load

the emf is easily measurable and the load flux is thus easily accessed, but in most AC motor controllers the flux vector used has to be estimated which is a difficult and usually parameter sensitive task. Thus the main parameter sensitivity is not in the DCC as such, but in the determination of the rotating reference frame which is as difficult with DCC as with sampled vector control.

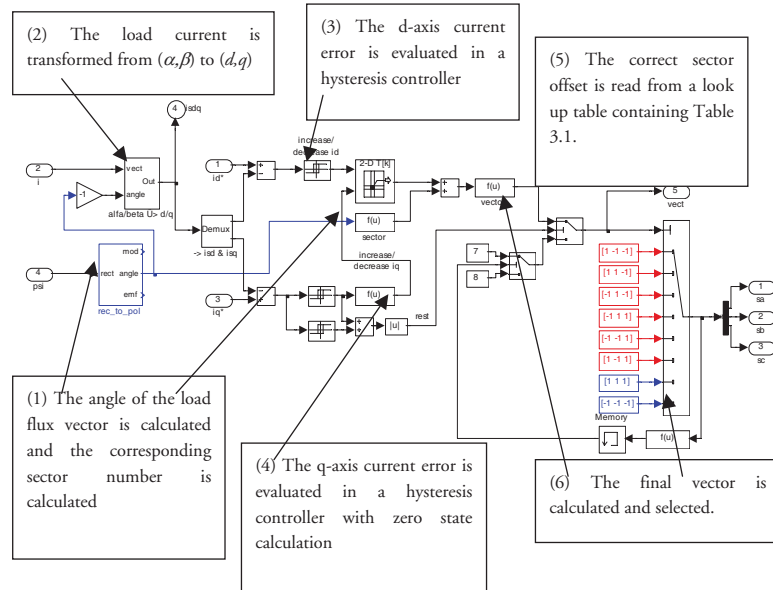


Figure 3.27: 3-phase DCC and it's components.

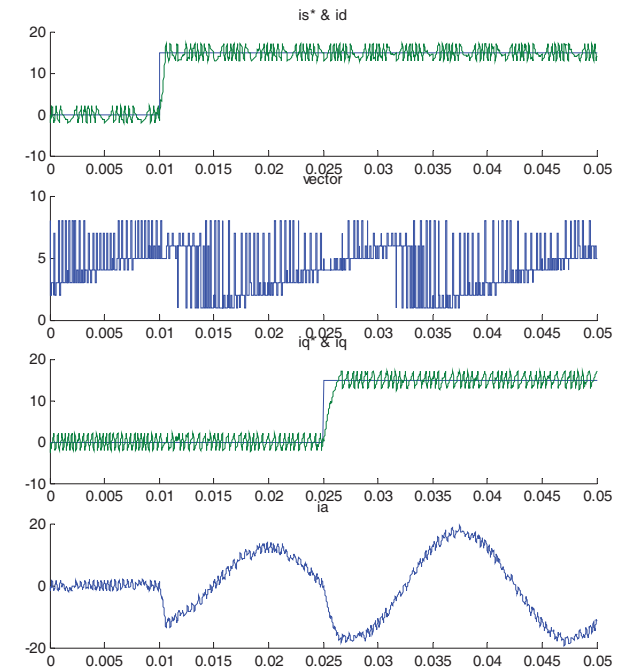


Figure 3.28: Step response of the 3-phase DCC method.